

SIMATS ENGINEERING

CO2 ASSESSMENT TOOL 2 – Scenario-Based Requirement Analysis

Course Code:	ITA0518	Course Title:	COMPUTER VISION
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
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Code of Conduct

I PRABHAVATHI R(192321101)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: PRABHAVATHI R

Annexure A – Scenario-Based Requirement Analysis

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

(a) Interpret the scenario and explain the cause of low contrast

A low contrast image is a grayscale image where the object and the background have very similar intensity values. As a result, the image looks dull, faded, and unclear, making it difficult to distinguish different objects.

Causes of low contrast:

- Poor or insufficient lighting during image capture.
- Limited camera sensor capability.
- Haze, fog, or shadows.

- Improper exposure settings.
- Pixel intensity values are concentrated within a small range.

(b) Identify the key issues affecting image visibility The main issues are:

- Poor distinction between object and background.
- Blurred or unclear edges.
- Important details are difficult to identify.
- Reduced image quality.
- Difficult for both human observation and computer vision algorithms to detect features accurately.

(c) Apply an appropriate enhancement technique Technique: Histogram Equalization Steps:

1. Read the grayscale image.
 2. Calculate the image histogram.
 3. Redistribute the intensity values over the full range (0–255).
 4. Generate the enhanced image. Result:
- Dark regions become clearer.
 - Bright regions become more visible.
 - Overall image contrast increases.
 - Objects are easier to distinguish from the background.

(d) Justify your choice

Histogram Equalization is the most suitable technique because:

- It spreads pixel intensity values across the entire grayscale range.
- It improves the visibility of hidden details.
- It enhances the contrast automatically without changing the image content.
- It is simple, fast, and widely used in medical imaging, satellite images, and computer vision applications.

Therefore, it is the best choice for enhancing a low contrast grayscale image.

(e) Clear and structured explanation

Problem:

The grayscale image has very low contrast, making the object and background appear almost the same.

Issues:

- Poor visibility
- Weak edges
- Hidden details
- Difficult object identification

Solution: Apply Histogram Equalization to redistribute pixel intensity values and improve contrast.

Outcome:

- Better brightness distribution
- Clear object boundaries
- Improved image quality

- Easier feature detection and analysis

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

(a) Interpret the scenario and explain the impact of uneven illumination

An **uneven illumination image** is one in which different parts of the image have different brightness levels. Some areas appear **too bright**, while others appear **too dark** due to **nonuniform lighting**.

Impact:

- Objects are not clearly visible.
- Important details may be hidden in dark regions.
- Bright regions may lose information due to overexposure.
- Image analysis and object detection become difficult.

(b) Identify the key factors causing intensity variation The main causes are:

- Non-uniform lighting conditions.
- Shadows cast by objects.
- Uneven light source distribution.
- Improper camera exposure settings.
- Reflections or glare from shiny surfaces.

(c) Apply a suitable enhancement method to normalize illumination

Steps:

1. Divide the image into small regions (tiles).
2. Apply histogram equalization to each region separately.
3. Limit excessive contrast amplification.
4. Combine all regions smoothly to produce the final image.

Result:

- Bright and dark areas become balanced.
- Local contrast is improved.
- Hidden details become visible.
- Overall image quality is enhanced.

(d) Justify your selected approach

CLAHE is the best choice because:

- It enhances **local contrast** instead of the whole image.
- It corrects uneven brightness across different regions.
- It prevents over-enhancement and noise amplification.
- It produces a natural-looking image with better visibility.

Therefore, CLAHE is more effective than standard Histogram Equalization for images with uneven illumination.

(e) Clear and structured explanation Problem:

The image contains bright and dark patches due to uneven lighting.

Issues:

- Uneven brightness
- Hidden details
- Poor visibility
- Difficult object detection

Solution:

Apply **CLAHE (Adaptive Histogram Equalization)** to normalize illumination and improve local contrast.

Outcome:

- Balanced brightness
- Clearer image
- Better visibility of details
- Improved object recognition uneven illumination.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

(a) Explain the scenario and characteristics of this noise

A **salt-and-pepper noise** image contains **random black (pepper) and white (salt) pixels** caused by errors during image acquisition, transmission, or scanning. In a scanned document, this noise reduces readability by adding unwanted dots.

Characteristics:

- Appears as random black and white pixels.
- Affects only some pixels.
- Makes text and image details difficult to read.
- Common in scanned documents and digital images.

(b) Identify the key issues impacting image

quality The main issues are:

- Reduced readability of the document.
- Loss of important image details.
- Presence of unwanted black and white dots.
- Difficulty in text recognition (OCR).
- Poor visual quality.

(c) Apply an appropriate filtering technique

Technique: Median Filter Steps:

1. Select a small neighborhood (e.g., 3×3 window).
2. Arrange the pixel values in ascending order.
3. Replace the center pixel with the **median** value.
4. Repeat for the entire image. **Result:**
 - Removes salt-and-pepper noise.
 - Preserves image edges.
 - Produces a smoother and clearer image.

(d) Justify why this method is more suitable than alternatives The **Median Filter** is the best choice because:

- It effectively removes salt-and-pepper noise.
- It preserves edges better than the Mean (Average) Filter.
- It does not blur important details.
- It improves readability while maintaining image quality.

Therefore, the Median Filter is more suitable than average or Gaussian filters for this type of noise.

(e) Clear and structured explanation Problem:

The scanned document contains random black and white pixels (salt-and-pepper noise), making it difficult to read.

Issues:

- Poor readability
- Random black and white dots
- Loss of image quality
- Difficult text recognition

Solution:

Apply a **Median Filter** to remove the noise while preserving edges.

Outcome:

- Noise-free image
- Clear and readable document
- Preserved text edges
- Improved image quality

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpret the scenario and explain why edges are blurred

The image has been processed to remove noise, but the filtering has made it **overly smooth**. As a result, **edges and fine details become blurred** because the filter averages neighboring pixel values.

Why edges are blurred:

- Strong smoothing filters remove both **noise and important edge information**.
- High-frequency components (edges) are reduced along with the noise.
- Excessive filtering causes loss of sharpness.

(b) Identify key issues related to smoothing and edge loss

The main issues are:

- Blurred object boundaries.
 - Loss of fine details.
 - Reduced image sharpness.
 - Difficulty in object detection and segmentation.
 - Lower overall image quality.
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(c) Apply an appropriate technique to restore edges

Technique: Sharpening using Unsharp Masking (High-Boost Filtering) Steps:

1. Blur the original image slightly.
2. Subtract the blurred image from the original to obtain edge details.
3. Add the edge details back to the original image.
4. Obtain a sharper image with restored edges.

Result:

- Sharper edges.
 - Better object boundaries.
 - Improved image clarity without significantly increasing noise.
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(d) Justify your decision

Unsharp Masking is the best choice because:

- It enhances edges while preserving most image details.
- It restores sharpness lost during smoothing.
- It improves visual quality without greatly amplifying noise.
- It is widely used in image enhancement and computer vision.

Therefore, it is more suitable than applying additional smoothing, which would blur the image further.

(e) Clear and structured explanation Problem:

Noise removal has made the image too smooth, causing blurred edges.

Issues:

- Blurred edges
- Loss of fine details
- Reduced sharpness
- Poor object detection

Solution:

Apply **Unsharp Masking (High-Boost Filtering)** to enhance and restore edges.

Outcome:

- Sharper edges
- Clear object boundaries
- Improved image quality
- Better feature detection

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

(a) Interpret the scenario and describe high-frequency noise

A **satellite image** contains **high-frequency noise**, which appears as fine random variations in pixel intensity. This noise makes the image look grainy and reduces the visibility of important features such as roads, rivers, buildings, and vegetation.

Characteristics of high-frequency noise:

- Appears as fine-grained random pixels.
 - Affects image sharpness and clarity.
 - Usually caused by sensor noise, transmission errors, or atmospheric disturbances.
 - Mainly exists in the **high-frequency components** of the image.
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(b) Identify the key issues affecting image clarity The main issues are:

- Grainy appearance of the image.
- Reduced visibility of important features.
- Difficulty in detecting edges and objects.
- Lower image quality.
- Reduced accuracy in image analysis and classification.

(c) Apply an appropriate frequency domain filtering method

Technique: Low-Pass Filter (LPF) in the Frequency Domain (Ideal, Butterworth, or Gaussian Low-Pass Filter) Steps:

1. Convert the image to the frequency domain using the **Fourier Transform (FFT)**.
2. Apply a **Low-Pass Filter** to remove high-frequency noise.
3. Preserve the low-frequency components containing the main image information.
4. Apply the **Inverse Fourier Transform (IFFT)** to obtain the enhanced image.

Result:

- High-frequency noise is reduced.
- Important image features become clearer.
- Overall image quality improves.

(d) Justify the choice of filter

A **Low-Pass Filter** is the best choice because:

- High-frequency noise is concentrated in the high-frequency region.
- The filter removes unwanted high-frequency components while preserving the main image content.
- It produces a smoother and clearer image.
- It is widely used in satellite and medical image processing.

Among LPFs, the **Gaussian Low-Pass Filter** is often preferred because it reduces noise smoothly and introduces fewer artifacts than an Ideal Low-Pass Filter.

(e) Clear and organized explanation Problem:

The satellite image contains fine-grained high-frequency noise, reducing feature visibility.

Issues:

- Grainy appearance
- Poor feature visibility
- Difficult edge detection
- Reduced image clarity

Solution:

Apply a **Gaussian Low-Pass Filter** in the frequency domain to suppress high-frequency noise.

Outcome:

- Reduced noise
- Improved image clarity
- Better visibility of roads, buildings, and other features
- Enhanced image quality

6. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

(a) Interpret the scenario and explain the importance

of boundary detection In a **medical image**, different tissues or organs must be separated accurately.

Boundary detection identifies the edges between tissues, helping doctors clearly distinguish normal and abnormal regions.

Importance:

- Identifies the boundaries of tissues and organs.
- Helps detect tumors or abnormal structures.
- Improves image analysis and diagnosis.
- Supports accurate image segmentation.

(b) Identify key challenges in detecting boundaries The

main challenges are:

- Noise in the medical image.
- Low contrast between tissues.
- Blurred or weak edges.
- Uneven illumination.
- Complex tissue structures.

(c) Apply a suitable edge detection method Technique:

Canny Edge Detection Steps:

1. Apply a Gaussian filter to remove noise.
2. Calculate image gradients to detect edge strength.
3. Perform non-maximum suppression to thin the edges.
4. Apply double thresholding.
5. Use edge tracking by hysteresis to obtain the final edges.

Result:

- Clear and thin boundaries.
- Reduced false edges.
- Accurate detection of tissue boundaries.

(d) Justify your selection

Canny Edge Detection is the best choice because:

- It removes noise before detecting edges.
- It detects both strong and weak edges accurately.
- It produces thin and continuous boundaries.
- It reduces false edge detection.
- It is widely used in medical image processing because of its high accuracy.

Therefore, Canny Edge Detection is more effective than simple edge detectors such as Sobel or Prewitt for medical images.

(e) Clear and systematic explanation Problem:

The boundaries between tissues in a medical image are difficult to identify.

Issues:

- Noise

- Low contrast
- Weak or blurred edges
- Difficult tissue separation **Solution:**

Apply **Canny Edge Detection** to accurately identify tissue boundaries.

Outcome:

- Clear and continuous edges
- Better tissue segmentation
- Improved medical diagnosis
- Enhanced image quality

7. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity. **(a)**

Interpret the scenario and explain segmentation requirements

The image contains **objects that are clearly different from the background based on pixel intensity**. This means the object pixels have noticeably higher or lower intensity values than the background, making them easy to separate.

Segmentation requirement:

- Divide the image into **object** and **background**.
- Separate the region of interest accurately.
- Produce a binary image for further analysis.

(b) Identify key features enabling separation The main features are:

- Clear difference in pixel intensity.
- High contrast between object and background.
- Well-defined object boundaries.
- Minimal noise in the image.

(c) Apply a suitable segmentation technique

Technique: Thresholding (Global Thresholding)

Steps:

1. Select a threshold value (**T**).
2. Compare each pixel intensity with **T**.
3. If the pixel value $\geq T$, classify it as the **object**.
4. If the pixel value $< T$, classify it as the **background**.
5. Generate a binary image.

Result:

- Objects are separated from the background.
- A clear binary segmented image is obtained.

(d) Justify your decision

Global Thresholding is the best choice because:

- The object and background have a clear intensity difference.
- It is simple, fast, and easy to implement.
- It provides accurate segmentation when lighting is uniform.
- It requires low computational cost.

Therefore, Global Thresholding is more suitable than complex segmentation methods for this scenario.

(e) Clear presentation Problem:

The objects are clearly distinguishable from the background using intensity values.

Issues:

- Need accurate object-background separation.
- Preserve object boundaries.
- Minimize misclassification.

Solution:

Apply **Global Thresholding** to convert the grayscale image into a binary image.

Outcome:

- Clear object extraction.
- Accurate segmentation.
- Improved image analysis.
- Easy feature detection.

8. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

(a) Interpret the scenario and explain segmentation challenges

The image contains **multiple objects or regions with similar (overlapping) intensity values**. Because the object and background do not have clear intensity differences, simple thresholding cannot separate them accurately.

Segmentation challenges:

- Objects have similar brightness.
- Boundaries between regions are unclear.
- Multiple regions must be identified correctly.

(b) Identify key issues such as intensity overlap The main issues are:

- Overlapping intensity values.
- Poor contrast between regions.
- Blurred or unclear boundaries.
- Noise affecting segmentation.
- Difficulty in separating adjacent objects.

(c) Apply an appropriate advanced segmentation method **Technique: Watershed Segmentation Steps:**

1. Preprocess the image to reduce noise.
2. Compute the gradient image.
3. Identify foreground and background markers.
4. Apply the **Watershed algorithm**.
5. Separate the image into different regions based on the detected boundaries.

Result:

- Multiple regions are segmented accurately.

- Touching or overlapping objects are separated.
- Clear region boundaries are obtained.

(d) Justify how the method handles complexity

Watershed Segmentation is the best choice because:

- It uses gradient information instead of only intensity values.
- It can separate touching or overlapping objects.
- It detects boundaries accurately even when intensity values are similar.
- It is widely used in medical and satellite image segmentation.

Therefore, Watershed Segmentation is more suitable than simple thresholding for complex images.

(e) Structured and clear explanation Problem:

The image contains multiple regions with overlapping intensity values.

Issues:

- Intensity overlap
- Unclear boundaries
- Noise
- Difficult object separation

Solution:

Apply **Watershed Segmentation** to separate regions using gradient information and markers.

Outcome:

- Accurate segmentation
- Clear region boundaries
- Separation of overlapping objects
- Improved image analysis

9. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

(a) Interpret the scenario and explain why edges are broken

After edge detection, the object boundaries appear **discontinuous (broken)** instead of forming complete outlines. This happens because weak edge pixels are not detected or are removed during processing.

Reasons for broken edges:

- Noise in the image.
- Weak or low-contrast edges.
- Improper threshold values.
- Excessive smoothing before edge detection.

(b) Identify key issues affecting boundary continuity The main issues

are:

- Discontinuous object boundaries.
 - Missing edge pixels.
 - Incomplete object shapes.
 - Difficulty in object detection and segmentation.
 - Reduced image analysis accuracy.
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(c) Apply a suitable method to improve edge connectivity Technique:

Morphological Closing (Dilation followed by Erosion) Steps:

1. Detect edges using an edge detector (e.g., Canny).
 2. Apply **Dilation** to connect nearby broken edges.
 3. Apply **Erosion** to restore the original edge thickness.
 4. Obtain continuous object boundaries. **Result:**
 - Broken edges are connected.
 - Continuous boundaries are formed.
 - Object shapes become complete.
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(d) Justify your approach

Morphological Closing is the best choice because:

- It connects small gaps between edge pixels.
 - It preserves the overall object shape.
 - It improves boundary continuity without significantly changing the image.
 - It enhances segmentation and object recognition.
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(e) Clear explanation Problem:

Object boundaries are broken after edge detection.

Issues:

- Discontinuous edges
- Missing boundary pixels
- Incomplete object shapes
- Difficult object detection **Solution:**

Apply **Morphological Closing (Dilation + Erosion)** to connect broken edges.

Outcome:

- Continuous boundaries
- Complete object shapes
- Improved segmentation
- Better image analysis

10. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) Interpret the scenario and define the combined requirement

An **industrial inspection system** must detect defects accurately. The image contains noise, but important edges must be preserved because defects are usually identified from object boundaries.

Combined requirement:

- Remove image noise.
 - Preserve important edges.
 - Improve defect detection accuracy.
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(b) Identify key issues involving noise and edge preservation The main issues are:

- Noise hides defects.
- Excessive smoothing blurs edges.

- Weak edges may disappear.
- Reduced defect detection accuracy.

(c) Apply an appropriate sequence of techniques

Technique Sequence:

1. **Median Filter** – Remove salt-and-pepper noise while preserving edges.
2. **Canny Edge Detection** – Detect clear and accurate edges.
3. *(Optional)* **Morphological Closing** – Connect small gaps in detected edges.

Result:

- Noise is removed.
- Important edges are preserved.
- Defects become easier to detect.

(d) Justify the order and selection

The selected order is appropriate because:

- **Median Filter** removes noise first without blurring edges.
- **Canny Edge Detection** works better on a clean image and accurately detects boundaries.
- **Morphological Closing** connects any broken edges after detection.
- This sequence improves defect detection accuracy while maintaining image quality.

(e) Clear and structured explanation Problem:

The inspection system requires both noise removal and edge preservation for accurate defect detection.

Issues:

- Noise affects image quality.
- Edge blurring reduces defect visibility.
- Difficult defect detection. **Solution:**
 1. Median Filter
 2. Canny Edge Detection
 3. Morphological Closing (if needed)
- **Outcome:**
 - Noise-free image
 - Preserved edges
 - Clear defect boundaries
 - Improved inspection accuracy

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Understand ing of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identificati on of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Applicatio n of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand